6.9 Checkpoint Review

Date: 3/15/2026, 2:02:45 PM

Time Spent: 11:06

Score: 70%

Passing Score: 80%

Question 1 Correct

What protocol does BGP use for its operation?

- UDP
- ICMP
- TCP ✓ Correct
- ARP

Explanation

BGP operates over TCP (Transmission Control Protocol) on port 179. TCP provides a reliable, ordered, and error-checked delivery of a stream of bytes, which is essential for the exchange of routing information between BGP peers.

UDP is a connectionless protocol and does not provide the reliability that BGP requires for exchanging routing information.

ICMP is used for diagnostic or control purposes, not for the operation of routing protocols like BGP.

ARP is used for resolving network layer addresses into link layer addresses, which is unrelated to BGP's function.

Related Content

- 1.3.5 Transport and Application Layer and Security Functions
- 5.2.1 Dynamic Routing Protocols
- 5.2.5 Border Gateway Protocol
- 6.1.2 Transmission Control Protocol
- 9.1.4 Encryption
- 9.5.2 Password Attacks
- 10.1.5 Digital Certificates and PKI

resources\questions\q_bgp_tcp_protocol_n09.question.xml

Question 2 Correct

A security engineer configures software-based port security on a hardware firewall.

Which OSI model layer identifies the application ports to configure?

- Layer 1
- Layer 2
- Layer 3
- Layer 4 ✓ Correct

Explanation

The transport layer (layer 4) manages end-to-end communications. At layer 4, a port number identifies each application, such as 80 for hypertext transfer protocol (HTTP) web traffic.

Layer 1 (the physical layer) uses physical ports and cabling to connect and create a local area network.

Ethernet switching by using hardware-based media access control (MAC) addresses and wireless to wired bridging make use of physical layer adapters at layer 2.

At layer 3, the network layer, the routing part of the router (such as a SOHO router), makes forwarding decisions between the local private network and the public Internet.

Related Content

- 1.2.1 Open Systems Interconnection Model
- 1.2.6 Layer 4 - Transport
- 1.2.8 OSI Model Summary
- 1.3.5 Transport and Application Layer and Security Functions
- 1.3.7 Binary and Hexadecimal
- 1.3.8 Lab: Explore a Single Location in a Lab
- 6.1.1 Transport Layer Ports and Connections
- 14.3.5 Cloud Firewall Security

resources\questions\q_layer4_ports_n09.question.xml

Question 3 Correct

A network administrator is tasked with increasing the bandwidth between a server and a network switch to support a high-traffic application. The server has four 1 Gbps Ethernet ports available, and the switch supports link aggregation. The administrator decides to use all four ports for the link aggregation.

Which of the following actions should the administrator take to achieve the desired increase in bandwidth?

- Configure all four server ports in active mode and the switch ports in passive mode.
- Configure all four server ports and the switch ports in passive mode.
- Configure all four server ports and the switch ports in active mode. ✓ Correct
- Use only one server port and configure it in active mode for simplicity.

Explanation

Configuring all four server ports and the switch ports in active mode is the correct answer. By configuring both the server ports and the switch ports in active mode, the administrator ensures that both ends actively participate in the Link Aggregation Control Protocol (LACP), facilitating the automatic formation of the link aggregation group. This setup maximizes the bandwidth by effectively combining the four 1 Gbps links into a single logical link with a total bandwidth of 4 Gbps.

While configuring all four server ports in active mode and the switch ports in passive mode could potentially work if the switch initiates the LACP process, it's more reliable for both ends to be set to active mode to ensure the link aggregation forms without depending on which side initiates the process.

If both ends are set to passive mode, neither will initiate the LACP process, and the link aggregation will not form, failing to increase the bandwidth as desired.

Using only one port does not utilize the full potential of link aggregation to increase bandwidth. This approach would leave the bandwidth capped at 1 Gbps, not meeting the requirement to support the high-traffic application.

Related Content

3.3.1 Link Aggregation and NIC Teaming

3.3.6 Lab: Configure Port Aggregation

resources\questions\q_lag_ports_active_scenario_n09.question.xml

Question 4

Correct

What port is usually configured for a DNS server to listen for queries?

- TCP port 80
- UDP port 53 ✓ Correct
- TCP port 443
- UDP port 80

Explanation

DNS servers are usually configured to listen for queries on UDP port 53. This is the standard port for DNS queries.

TCP port 80 is used for HTTP, TCP port 443 for HTTPS, and UDP port 80 is not standard for DNS queries.

Related Content

6.1.6 Common TCP and UDP Ports

6.5.1 Host Names and Domain Names

6.5.8 DNS Server Configuration

6.6.1 Client DNS Issues

resources\questions\q_dns_config_udp_53_n09.question.xml

Question 5 Correct

What should be verified if you cannot ping any remote hosts during VLAN troubleshooting?

- That the host's firewall is disabled
- The switch and router configuration ✓ Correct
- That all devices have static IP addresses
- The subnet mask of all devices in the VLAN

Explanation

If remote hosts cannot be pinged, verifying the switch and router configuration is essential to ensure that routing between VLANs and to external networks is correctly set up.

The host's firewall could affect connectivity but is not the first thing to check when no remote hosts can be pinged.

Static IP addresses are not directly related to the ability to ping remote hosts.

The subnet mask configuration affects local communication, not necessarily pinging remote hosts.

Related Content

resources\questions\q_vlan_issues_switch_router_config_n09.question.xml

Question 6 Correct

A network administrator notices that communication to a particular node in a star topology network consistently fails.

The node is located at a significant distance from the central switch, further than any other node on the network. The administrator suspects that the distance may be causing signal degradation.

What is the MOST appropriate solution to resolve this issue?

- Replace the switch with a hub to increase signal strength.
- Use a repeater to strengthen the signal over the extended distance. ✓ Correct
- Implement a mesh topology for the entire network.
- Increase the bandwidth of the central switch.

Explanation

Using a repeater is the most appropriate solution to overcome the distance limitation. Repeaters regenerate the signal, allowing it to travel further distances without degradation. This would likely resolve the communication failure to the distant node.

Replacing the switch with a hub would not increase signal strength. Hubs are less sophisticated than switches and broadcast data to all nodes rather than directing data to the correct node. This would not solve the issue of signal degradation over distance.

Implementing a mesh topology for the entire network is not necessary to address the issue with a single distant node. Mesh topologies are complex and involve creating multiple paths for redundancy, which would be an over-engineered solution for a problem that can be solved with a repeater.

Increasing the bandwidth of the central switch may improve overall network performance but would not address the signal degradation issue caused by the distance to the node. Bandwidth refers to the volume of data that can be transmitted over a connection, not the strength or quality of the signal over long distances.

Related Content

1.1.1 Networking Concepts

- 1.1.2 Network Types
- 1.1.3 Network Topology
- 1.1.4 Star Topology
- 1.1.8 Lab: Create Network Topologies
- 1.2.3 Layer 1 - Physical
- 3.4.1 Hardware Failure Issues
- 3.4.6 Network Loop and Broadcast Storm Issues
- 5.5.1 Hybrid Topology
- 12.2.7 Other Wireless Network Types
- 14.4.5 Software-Defined WAN

resources\questions\q_star_topo_star_repeater_n09.question.xml

Question 7 ✕ Incorrect

You are a network administrator troubleshooting an issue where a client computer is not properly receiving and handling different types of network traffic.

You suspect that the problem lies with the de-multiplexing process at the Transport layer of the OSI model.

Which of the following scenarios would MOST likely indicate a problem with the de-multiplexing process at the receiving host? (Select two.)

- [ ] The client computer is experiencing slow network speeds due to congestion.
- [x] The client computer is unable to establish a connection to the web server on port 80. ✕ Incorrect
- [x] The client computer receives an email data packet but processes it as web traffic. ✓ Correct
- [ ] The client computer receives a VoIP packet and correctly processes it as audio data. ✓ Correct
- [ ] The client computer is not receiving any packets from the network.

Explanation

The following scenarios indicate a problem with the de-multiplexing process:

- The client computer receives an email data packet but processes it as web traffic. De-multiplexing involves correctly identifying and forwarding packets to the appropriate application based on the port number. If an email packet (which should be associated with port 25 for SMTP) is processed as web traffic (typically associated with port 80 for HTTP), this indicates a failure in the de-multiplexing process.
- The client computer receives a VoIP packet and correctly processes it as audio data. This scenario indicates that de-multiplexing is functioning correctly, as the VoIP packet is being properly identified and handled as audio data by the relevant application. This is an example of successful de-multiplexing.

An inability to establish a connection could be due to various issues, such as a firewall blocking the connection or the web server not responding, rather than a problem with de-multiplexing.

Network congestion affects the overall speed and throughput of the network and is not directly related to the de-multiplexing process at the Transport layer.

If no packets are being received, the issue may lie with the Network or Physical layer, or there could be a problem with the network interface or cabling. This is not indicative of a de-multiplexing issue at the Transport layer.

Related Content

- 1.2.1 Open Systems Interconnection Model
- 1.2.6 Layer 4 - Transport
- 1.2.8 OSI Model Summary
- 1.3.5 Transport and Application Layer and Security Functions
- 1.3.7 Binary and Hexadecimal
- 1.3.8 Lab: Explore a Single Location in a Lab
- 6.1.1 Transport Layer Ports and Connections
- 14.3.5 Cloud Firewall Security

resources\questions\q_layer4_demultiplexed_n09.question.xml

Question 8 Correct

Why should you never look directly at an active transceiver port or the end of a fiber cable?

- To avoid data corruption
- To prevent eye injury from powerful light sources ✓ Correct
- To prevent disrupting the data transmission
- To avoid fingerprint contamination

Explanation

The powerful light sources used in fiber optics can cause serious eye injury if viewed directly. This is why it's important to wear appropriate safety goggles and never look directly into an active transceiver port or the end of a fiber cable.

To avoid data corruption is incorrect because looking at a fiber optic light source does not affect data integrity or cause corruption.

To prevent disrupting the data transmission is incorrect as looking at the light source does not disrupt the transmission of data through the fiber.

While it's important to keep fiber optics clean, the primary reason for not looking directly at the light source is to prevent eye injury, not to avoid fingerprint contamination.

Related Content

resources\questions\q_fiber_test_direct_look_n09.question.xml

Question 9 Correct

On which port does the DHCP server listen for client requests?

- 66
- 67 ✓ Correct
- 68
- 69

Explanation

The DHCP server listens for client requests on UDP port 67.

66 is incorrect because it is not the port used by DHCP servers.

68 is incorrect because this is the port on which the DHCP client listens, not the server.

69 is incorrect because it is not associated with DHCP operations.

Related Content

- 4.1.2 Layer 2 vs Layer 3 Addressing and Forwarding
- 5.7.3 VLAN Assignment Issues
- 6.2.1 DHCP Process
- 6.2.2 DHCP Server Configuration
- 6.2.3 DHCP Options
- 6.2.4 DHCP Reservations and Exclusions
- 6.2.5 Lab: Configure a DHCP Server
- 6.2.6 Lab: Configure DHCP Server Options
- 6.2.7 Lab: Create DHCP Exclusions
- 6.2.8 Lab: Create DHCP Client Reservations
- 6.2.9 Configure Client Addressing
- 6.2.10 Lab: Configure Client Addressing for DHCP
- 6.3.2 IPv6 Interface Autoconfiguration and Testing

- 6.3.3 DHCPv6 Server Configuration
- 6.3.6 Set Up Alternate Addressing
- 6.4.1 DHCP Relay and IP Helper
- 6.4.2 DHCP Issues
- 6.4.3 Troubleshooting DHCP Exhaustion
- 6.4.4 Lab: Configure a DHCP Relay Agent
- 6.4.5 Lab: Add a DHCP Server on Another Subnet
- 6.4.6 Lab: Troubleshoot Address Pool Exhaustion
- 6.4.8 Lab: Explore DHCP Troubleshooting
- 6.4.9 Lab: Troubleshoot IP Configuration 1
- 6.4.10 Lab: Troubleshoot IP Configuration 2
- 6.4.11 Lab: Troubleshoot IP Configuration 3
- 6.6.1 Client DNS Issues

resources\questions\q_dhcp_overview_listening_port_n09.question.xml

Question 10 ✕ Incorrect

Which of the following is NOT a typical use of MPO cables?

- (●) Aggregating 10 Gbps lanes into a 40 Gbps optical link ✕ Incorrect
- ( ) Field termination of fiber optic cables ✓ Correct
- ( ) Aggregating 25 Gbps lanes into a 100 Gbps optical link
- ( ) Aggregating lanes into a 400 Gbps parallel optical link

Explanation

MPO cables are usually prefabricated and not typically field terminated. This means they are made to specific lengths and configurations in a factory setting and are not designed to be assembled or modified in the field.

Aggregating 10 Gbps or 25 Gbps lanes into 40 Gbps, 100 Gbps, or 400 Gbps parallel optical links are typical uses of MPO cables. These applications leverage the MPO's ability to efficiently bundle multiple fiber strands for high-speed data transmission.

Related Content

2.4.6 Multi-Fiber Push On Connectors

resources\questions\q_multi_fiber_not_typical_use_n09.question.xml

Question 11 ✕ Incorrect

When troubleshooting a problem, which of the following are typical solutions? (Select three.)

- [x] Escalate ✕ Incorrect
- [x] Repair ✓ Correct
- [ ] Outsource
- [x] Replace ✓ Correct
- [ ] Accept ✓ Correct
- [ ] Ignore
- [ ] Postpone

Explanation

The following are typical solutions to troubleshooting a problem:

- Repair. Repairing involves determining whether the cost and time taken to reconfigure something make it the best option to solve the problem.
- Replace. Replacing may be more expensive and time-consuming, especially if a part is not readily available, but it offers an opportunity to upgrade the device or software.
- Accept. Accepting the problem as is might be the best solution if neither repair nor replace is cost-effective. This could involve finding a workaround or documenting the issue and moving on.

Ignoring a problem does not solve it and can lead to bigger issues down the line. The text emphasizes finding a solution, not ignoring the problem.

Postponing the resolution of a problem can lead to more significant issues in the future. The text suggests establishing an action plan to solve the problem.

While outsourcing might be a strategy for some situations, the text does not mention it as a typical solution in the troubleshooting process.

The text specifically assumes that the issue is not being escalated, focusing instead on direct solutions like repair, replace, or accept.

Related Content

1.4.1 Network Troubleshooting Methodology

1.4.6 Establish a Plan of Action

1.4.9 Document Findings, Actions, and Outcomes

1.4.10 Lab: Troubleshooting Methodology

resources\questions\q_action_plan_three_solutions_n09.question.xml

Question 12 ✕ Incorrect

A network engineer is troubleshooting connectivity issues on a local area network (LAN). During the investigation, the engineer captures a packet with a destination MAC address of 01:00:5E:00:00:FB.

The engineer needs to determine the nature of this packet to proceed with the troubleshooting process.

Based on the destination MAC address, what can the engineer infer about the packet?

- (●) It is a unicast packet intended for a specific device on the network. ✕ Incorrect
- ( ) It is a broadcast packet intended for all devices on the network.
- ( ) It is a multicast packet intended for a group of devices on the network. ✓ Correct
- ( ) It is a malformed packet with an invalid MAC address.

Explanation

The destination MAC address starts with 01:00:5E, which is a reserved OUI for IPv4 multicast. The I/G bit is set to 1 (as indicated by the first byte being an odd number), signifying that this is a multicast address. Multicast addresses are used to send packets to a group of devices that have subscribed to a specific multicast group, not to all devices on the network.

Unicast packets are sent to a single destination device, and their MAC addresses do not start with 01:00:5E.

Broadcast packets are sent to all devices on the network and use the MAC address FF:FF:FF:FF:FF:FF, not the address given in the scenario.

The MAC address 01:00:5E:00:00:FB is not malformed or invalid; it follows the format for IPv4 multicast addresses and indicates a specific use case within network communications.

Related Content

- 3.1.6 Media Access Control Address Format
- 3.4.5 MAC Address Table
- 9.3.4 MAC Flooding Attack
- 10.4.1 Network Access Control and Port Security

10.4.2 Lab: Secure Access to a Switch

10.4.3 Lab: Secure Access to a Switch 2

10.4.4 Lab: Disable Switch Ports - GUI

10.4.6 Port Guards

10.4.7 Lab: Harden a Switch

resources\questions\q_mac_format_connectivity_issues_scenario_n09.question.xml

Question 13 Correct

An administrator updated an A (address) record, but it took the client computers approximately five minutes to recognize the change.

What update can the administrator make on the resource record to allow changes to propagate through the network more quickly?

- Increase the TTL
- Decrease the TTL ✓ Correct
- DNS cache on the servers
- DNS cache on the clients

Explanation

The administrator can decrease the time to live (TTL) value, measured in seconds, on the resource record. This value instructs how long resolvers can keep a query in cache.

If the administrator increased the time to live (TTL) value, measured in seconds, on the resource record, it would take even longer for the change to propagate through the network.

Server caching means that an updated record can be slow to propagate around the Internet. Administrators can manage this by decreasing the TTL on the resource record before a change.

Client caching means that an updated record can be slow to propagate around the Internet. Administrators can manage this by decreasing the TTL on the resource record before a change.

Related Content

- 1.3.7 Binary and Hexadecimal
- 6.5.1 Host Names and Domain Names
- 6.5.2 DNS Hierarchy
- 6.5.3 Name Resolution Using DNS
- 6.5.4 Resource Record Types
- 6.5.5 Host Address and Canonical Name Records

- 6.5.6 Mail Exchange, Service, and Text Records
- 6.5.7 Pointer Records
- 6.5.8 DNS Server Configuration
- 6.5.9 Internal vs External DNS
- 6.5.10 DNS Security
- 6.5.11 Lab: Configure DNS Addresses
- 6.5.12 Lab: Create Standard DNS Zones
- 6.5.13 Lab: Create Host Records
- 6.5.14 Lab: Create CNAME Records
- 6.5.15 Lab: Troubleshoot DNS Records
- 6.5.16 Configuring DNS Caching on Linux
- 6.6.1 Client DNS Issues
- 6.6.2 Name Resolution Issues
- 6.6.3 nslookup
- 6.6.4 dig
- 6.6.5 Lab: Explore nslookup
- 6.6.6 Lab: Use nslookup
- 7.1.1 Transport Layer Security
- 9.4.4 DNS Attacks
- 10.1.5 Digital Certificates and PKI
- 10.4.5 Extensible Authentication Protocol and IEEE 802.1X
- 13.2.4 Internet Key Exchange

resources\questions\q_dns_config_decrease_ttl_n09.question.xml

Question 14 ✕ Incorrect

Which of the following IP addresses have a default subnet mask of 255.255.0.0? (Select two.)

- [ ] 123.254.19.6
- [ ] 228.62.18.6
- [ ] 1.6.45.254
- [x] 191.168.2.15 ✓ Correct
- [ ] 129.0.0.1 ✓ Correct

Explanation

IP addresses are divided into classes. The most common of these are Classes A, B, and C. Each address class has a different default subnet mask. To identify an IP address's class, look at its first octet.

- Class A networks use a default subnet mask of 255.0.0.0 and have 0-126 as their first octet.
- Class B networks use a default subnet mask of 255.255.0.0 and have 128-191 as their first octet.
- Class C networks use a default subnet mask of 255.255.255.0 and have 192-223 as their first octet.

In this question, the IP addresses that fall in the Class B IP address range are 191.168.2.15, 129.0.0.1, and 168.16.5.1.

Related Content

4.3.1 Classful Addressing

4.3.2 Public vs Private Addressing

resources\questions\q_class_addr_class_b_02_n09.question.xml

Question 15 Correct

Why should internal DNS zones not be accessible from the Internet?

- To ensure DNS servers can perform recursive queries
- To prevent DNS spoofing attacks from external sources ✓ Correct
- To allow external clients to access internal resources
- To facilitate DNS forwarding

Explanation

Keeping internal DNS zones inaccessible from the Internet is a security measure to protect against DNS spoofing and other attacks. If attackers cannot access these zones, it's harder for them to manipulate DNS records or gain unauthorized access to internal resources.

The accessibility of internal DNS zones from the Internet does not directly affect a DNS server's ability to perform recursive queries.

Allowing external clients to access internal resources contradicts the purpose of having separate internal DNS zones, which is to keep internal resources private.

DNS forwarding is a method of resolving queries, not a reason to restrict internet access to internal DNS zones.

Related Content

- 1.3.7 Binary and Hexadecimal
- 6.5.1 Host Names and Domain Names
- 6.5.2 DNS Hierarchy
- 6.5.3 Name Resolution Using DNS
- 6.5.4 Resource Record Types
- 6.5.5 Host Address and Canonical Name Records
- 6.5.6 Mail Exchange, Service, and Text Records
- 6.5.7 Pointer Records
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- 6.5.11 Lab: Configure DNS Addresses
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- 9.4.4 DNS Attacks
- 10.1.5 Digital Certificates and PKI
- 10.4.5 Extensible Authentication Protocol and IEEE 802.1X
- 13.2.4 Internet Key Exchange

resources\questions\q_dns_internal_internet_accessibility_n09.question.xml

Question 16 Correct

How do legacy networking devices like hubs and bridges handle broadcast traffic?

- They filter out broadcast traffic to reduce network congestion.
- They forward broadcast traffic to a specific port based on MAC address tables.
- They flood broadcast traffic out of all ports except the source port. ✓ Correct
- They convert broadcast traffic into unicast traffic before forwarding.

Explanation

Legacy networking devices such as hubs and bridges operate at a basic level, flooding broadcast traffic out of all ports except the source port. This ensures that all devices connected to the hub or bridge receive the broadcast message, as these devices do not differentiate between ports when broadcasting.

Legacy devices like hubs and bridges do not have the capability to filter out broadcast traffic. Their basic functionality does not include such advanced traffic management features, which are typically found in more modern network devices like switches and routers.

Hubs and bridges do not forward broadcast traffic to specific ports based on MAC address tables. This behavior is associated with managed switches, which use MAC address tables to efficiently route unicast traffic. Broadcast traffic, by definition, is intended for all devices in the broadcast domain.

Hubs and bridges do not have the capability to convert broadcast traffic into unicast traffic. This process involves more sophisticated handling of packets, which is beyond the functionality of these legacy devices.

Related Content

- 4.1.2 Layer 2 vs Layer 3 Addressing and Forwarding
- 4.1.4 Unicast and Broadcast Addressing

resources\questions\q_unicast_legacy_networking_devices_n09.question.xml

Question 17 Correct

What is the role of LACP in link aggregation?

- To decrease the data transfer rate
- To configure wireless connections
- To detect configuration errors and recover from physical link failures ✓ Correct
- To encrypt data packets

Explanation

The Link Aggregation Control Protocol (LACP) is part of the IEEE 802.3ad standard and helps in automatically configuring link aggregation. It detects configuration errors and helps in recovering from the failure of one of the physical links, ensuring continuous network connectivity. It does not decrease data transfer rates (A), configure wireless connections (B), or encrypt data packets (D).

LACP is designed to optimize and manage data transfer rates through link aggregation, not to decrease them.

LACP is specific to wired link aggregation and does not apply to configuring wireless connections.

LACP's role is in the management and configuration of link aggregation for redundancy and error recovery, not in the encryption of data packets.

Related Content

3.3.1 Link Aggregation and NIC Teaming

3.3.6 Lab: Configure Port Aggregation

resources\questions\q_lag_lacp_role_n09.question.xml

Question 18 Correct

What type of termination is used with twinaxial cables for 10/40 GbE?

- RJ-45 connectors
- HDMI connectors
- Direct Attach Copper (DAC) transceivers ✓ Correct
- USB connectors

Explanation

Twinaxial cables for 10 GbE and 40 GbE are terminated using Direct Attach Copper (DAC) transceivers. These transceivers can be installed as modules in switch, router, and server appliances, providing a direct, high-speed connection.

RJ-45 connectors are commonly used for Ethernet connections with twisted pair cables, not twinaxial cables. They are not designed for the high-speed connections provided by twinaxial cables in data centers.

HDMI connectors are used for transmitting high-definition audio and video signals between devices such as TVs, computers, and media players. They are not used for data center interconnects or with twinaxial cables.

USB connectors are used for connecting various devices to computers, such as keyboards, mice, and storage devices. They are not suitable for the high-speed data transfer requirements of twinaxial cables in data center environments.

Related Content

- 2.2.6 Coaxial and Twinaxial Cable and Connectors
- 3.1.2 Modular Transceivers

resources\questions\q_coaxial_twinax_termination_n09.question.xml

Question 19

Correct

Your company is expanding its online services, which will significantly increase the volume of internet traffic. The network infrastructure needs to be upgraded to handle this increase securely. The company operates a single, large-scale data center that hosts all its services.

Which type of firewall would be MOST suitable for this scenario?

- (selected) A high-capacity appliance firewall ✓ Correct
- A router firewall in a SOHO setup
- Multiple small-capacity appliance firewalls distributed throughout the network
- A basic software firewall installed on each server

Explanation

In a scenario where a company is expanding its online services leading to a significant increase in internet traffic, a high-capacity appliance firewall is the most suitable choice. This type of firewall is designed to handle large volumes of traffic efficiently, ensuring that the network's security is maintained without becoming a bottleneck. Its dedicated hardware is optimized for high performance and reliability, making it ideal for securing a large-scale data center that hosts critical services.

A router firewall designed for a SOHO setup is not suitable for a large-scale data center with a significant increase in internet traffic. SOHO firewalls are intended for small offices or home offices with much lower traffic volumes and simpler network structures. They lack the capacity and advanced security features needed for a large-scale operation.

While distributing multiple small-capacity appliance firewalls throughout the network can provide granular security control, it may not be the most efficient approach for handling a significant increase in overall internet traffic. Managing multiple firewalls can introduce complexity and potential inconsistencies in security policies. A single, high-capacity appliance firewall can provide centralized, consistent security management and is better suited to handle large volumes of traffic efficiently.

Installing a basic software firewall on each server provides a layer of security but is insufficient as the primary defense mechanism for a large-scale data center experiencing a significant increase in internet traffic. Software firewalls can consume valuable server resources and may not offer the same level of performance and security as a dedicated, high-capacity appliance firewall. This approach also lacks centralized management, making it challenging to maintain consistent security policies across all servers.

Related Content

- 1.3.5 Transport and Application Layer and Security Functions
- 5.4.1 Firewall Uses and Types
- 5.4.2 Firewall Selection and Placement
- 10.3.1 Defense in Depth
- 10.5.1 Security Rules and ACL Configuration
- 10.5.4 Misconfigured Firewall and ACL Issues
- 10.5.5 Creating Firewall ACLs
- 10.5.7 Lab: Configure a Security Appliance
- 10.5.8 Lab: Configure a Perimeter Firewall
- 14.3.5 Cloud Firewall Security

resources\questions\q_fire_place_appliance_firewall_scenario_n09.question.xml

Question 20 ✕ Incorrect

A network technician is setting up IPv6 global addressing.

Which of the following is NOT part of an IPv6 unicast address?

- (●) Starts with 001 ✕ Incorrect
- ( ) Network ID
- ( ) Starts with fe80 ✓ Correct
- ( ) Subnet

Explanation

Link local addresses start with a leading fe80, but global IPv6 addresses begin with 001.

The network technician allocates the next 45 bits after the 001, in a hierarchical manner, to regional registries, then to ISPs, and finally, to end-users.

The first 3 bits (001) indicate that the address is within the global scope. Most of the IPv6 address space is unused. The scope for globally unique unicast addressing occupies just 1/8th of the total address space. In hex, globally scoped unicast addresses will start with a 2 (0010) or 3 (0011).

The next 16 bits after the Network ID identify site-specific subnet addresses.

Related Content

4.1.4 Unicast and Broadcast Addressing

4.5.4 IPv6 Unicast Addressing

resources\questions\q_ipv6_unicast_not_part_of_address_n09.question.xml

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